# Module : 5 Electrical Machine

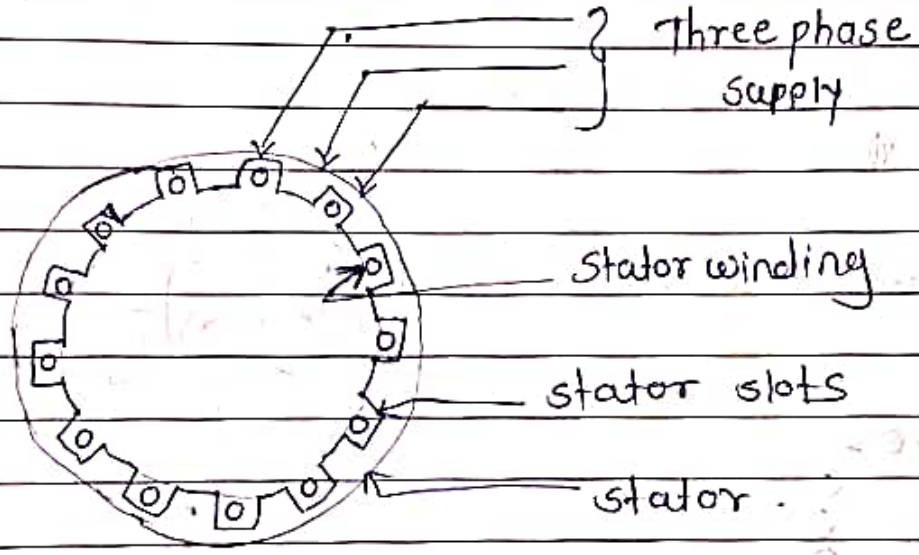
## (*) Principle of operation of 3-ϕ Induction motor & Construction of 3-ϕ Induction motor.

### Construction of 3-ϕ IM

> A 3-ϕ I.M consist of two parts : the stator & the rotor.

> The stator is the sationary part & the rotor is the rotating part.

Stator




1> stator is an outer 'stationary' hollow cylindrical structure made of laminated sheet steel.

2> The inner periphery of the stator is slotted to receive the winding.

3> The stator carries the Three-phase winding and is fed from a 3-ϕ supply.

4) it is wound on definite number of pole is determined by the requirment of speed.

5> greater the number of pole, lesser is the speed and vice versa.

6) when the stator winding is fed from a three-phase supply, it produces a magnetic field, which is of constant magnitude but revolves at synchronous speed.

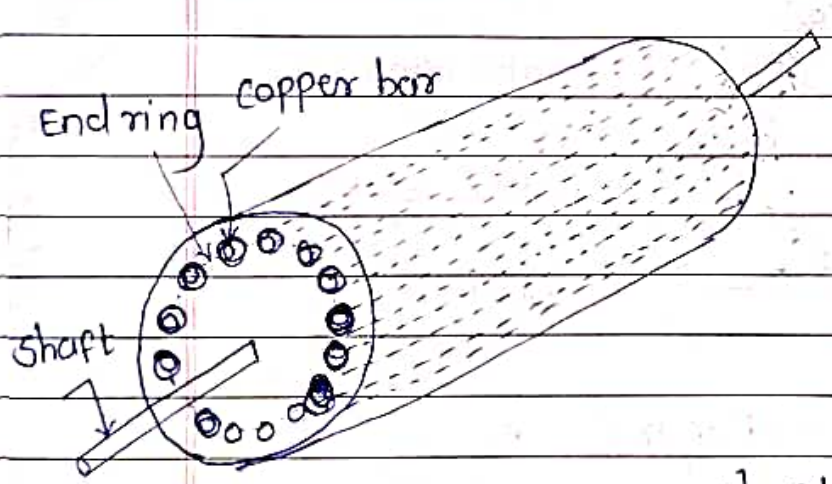
## Rotor

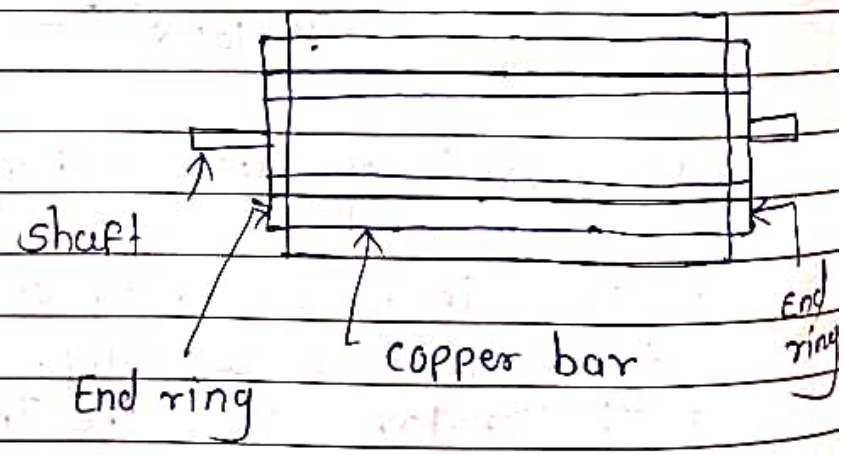
> Rotor is an inner cylindrical core mounted on the shaft.

> There are two types of rotor.

① squirrel cage Rotor
② slip ring rotor.

### ① Squirrel cage Rotor




1) This rotor has the simplest and most rugged / Rough construction.

2) It is consist of a cylindrical laminated core with slots parallel to the shaft axis.

3) each slot Contains one bar Conductor of alluminium or copper.

4) At the Both end of the rotor, the bar conductor are short circuited by heavy end rings of the same material.

5) The bar Conductors & end rings form a cage of the type that was once commonly used for keeping squirrels, hence it's name.

6) it should be noted that the rotor bars are permanently short circuited on themselves.

7) hence it is not possible to add any external resistance in series with the rotor circuit for starting purposes.

## ② Slip-ring rotor / Phase-wound rotor

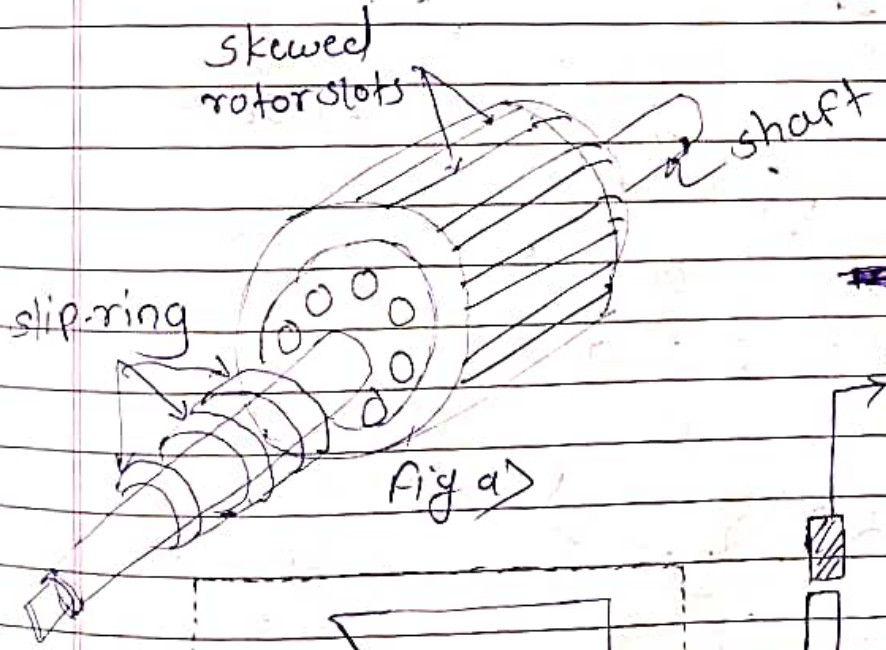
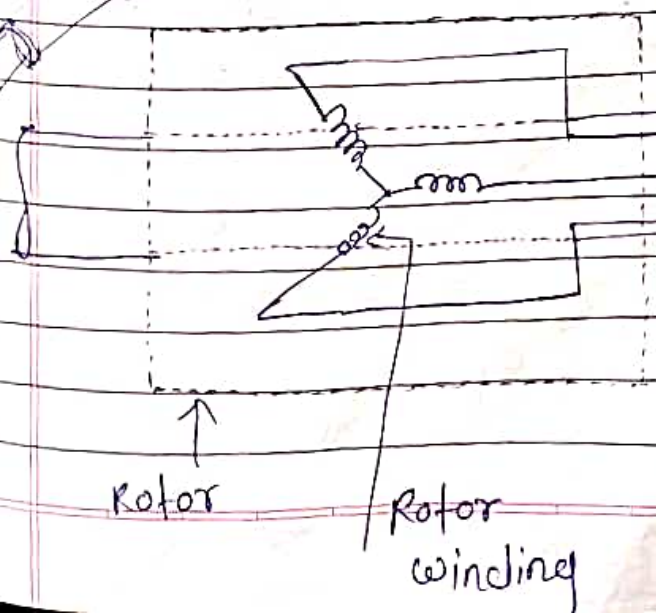



Fig a)



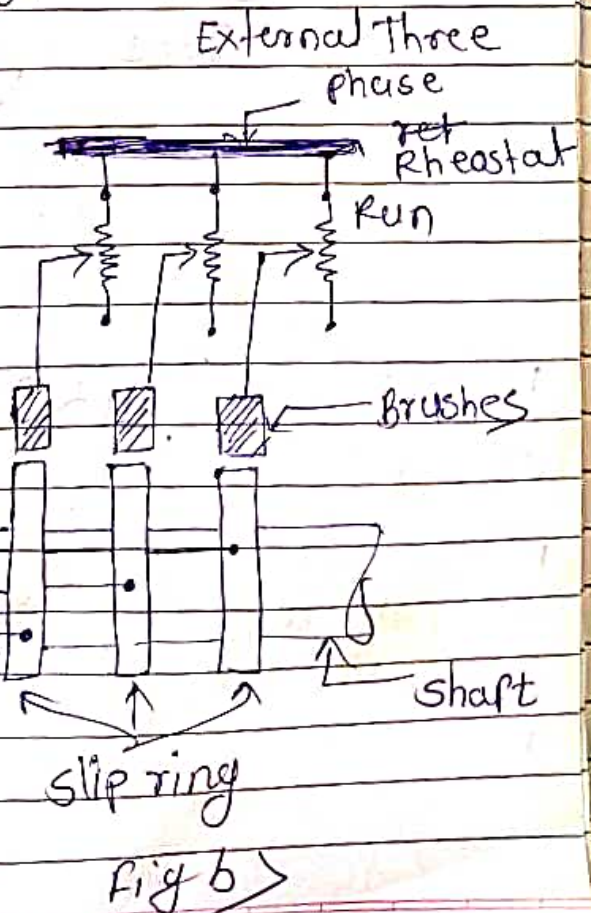
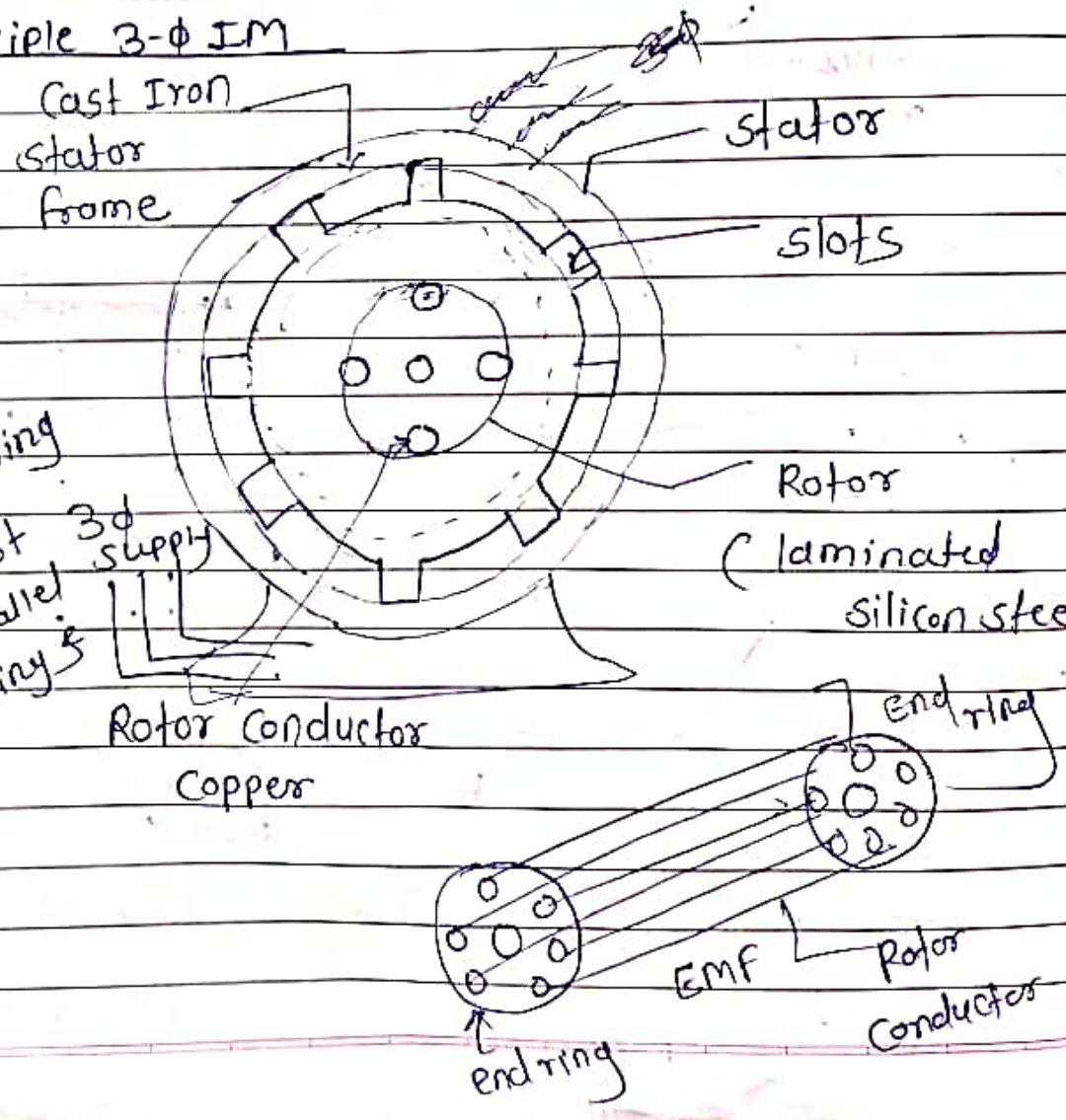




Fig b)

> This rotor consist of a slotted laminated core of sheet steel.
> A Three phase winding is placed in the rotor slots, having the same number of poles as the stator.
> The rotor winding is connected in star.
> slip ring mounted on the shaft with brushes resting on them shown in fig b>
> These brushes are externally connected to a Three-phase star-connected rheostat.
> Additional resistance in the rotor circuit during the starting period for increasing the starting torque of the motor.
> Also varying the external resistance, the speed of the motor can be controlled.

Working /principle 3-ϕ IM



① To gives the 3-ϕ supply to the stator winding. the stator winding establish the Rotating magnetic field to the inner portion of stator.

② The established Rotating magnetic field rotates by using the synchronous speed of the

Working principle

We give the 3-ϕ supply to the stator winding

↓

Rotating magnetic field established around the stator

↓

Rotating magnetic field rotates at synchronous speed

↓

Equation synchronous speed $N_S = \frac{120f}{P}$

P = no. of pole

f = frequency

$N_S$ = synchronous speed

(synchronous speed ⇒ speed of rotating magnetic field is called synchronous speed)

↓

Establishing rotating magnetic field cuts the Rotor bar ~~Conductor~~

↓

Rotor conductor induces emf from faradays law of electromagnetic induction

↓

since rotor conductor remain connect with end ring so rotor conductor from a complete circuit

Torque ⇒ torque is the measure of the force that can cause an object to rotate about an axis.

So the emf induced in the conductor, due to induced emf current circulates in the conductor of rot

↓

due to circulate current another magnetic field creates around the rotor cond$^r$

↓

as we known when current carrying conductor placed in magnetic field, then the torque is act in that conductor

↓

Here current circulateing along the rotor conductor and magnetic field present around the rotor conductor

↓

that's why rotor experiences a force

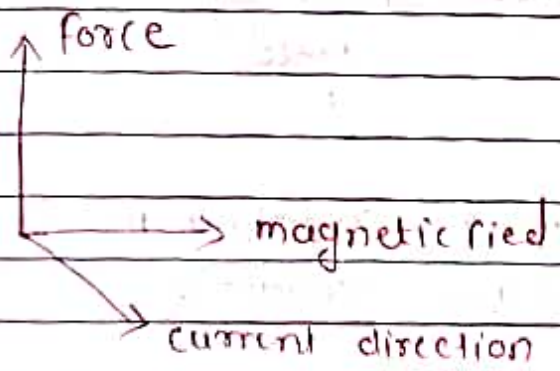
↓

when torque act tangentially on the rotor

↓

Rotor starts to rotate with rotating magnetic field

↓

direction of rotation determine by fleming left hand Rule.

fleming left hand Rule